

E-Commerce Analytics Platform

Revenue Optimization & Customer Intelligence

End-to-end analytics pipeline — MySQL · Python · Power BI

Project Overview

A fast-growing e-commerce company faced rising customer acquisition costs, a 68% cart abandonment rate, 22% product return rate, frequent inventory stockouts, and only 78% on-time delivery rate. This project builds a centralized E-Commerce Analytics Platform providing 360° visibility across customer behaviour, sales performance, marketing effectiveness, inventory movement, and logistics operations.

Metric	Value
Sales Records	50,000
Customers	5,000
Products	1,000
Phases Completed	8
ML Models Built	3

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Phase 1 — Database Design & ETL

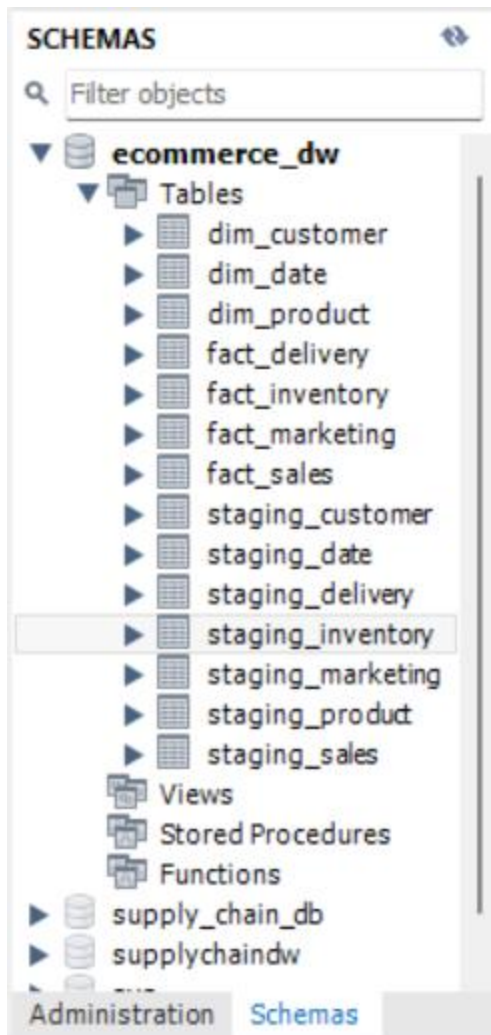
Star schema architecture · SCD Type 2 · 7 tables · MySQL Workbench

Architecture

Designed a star schema data warehouse with three central fact table and four dimension tables. Applied Slowly Changing Dimension Type 2 on dim_customer to track historical changes in customer attributes. The dim_date table enables time intelligence for MoM, YoY, and quarterly analysis.

Dataset Summary

Table	Source File	Rows	Key Columns
fact_sales	Sales.csv	50,000	Order_ID, Customer_Key, Product_Key, Net_Amount, Return_Flag
dim_customer	Customer.csv	5,000	Customer_Key, Region, Loyalty_Tier, CLV, Age_Band
dim_product	Product.csv	1,000	Product_Key, Category, Brand, Unit_Price, Cost_Price
fact_delivery	Delivery.csv	50,000	Delivery_ID, Partner_Name, Actual_Delivery_Days, Delivery_Status
fact_inventory	Inventory.csv	13,250	Product_Key, Warehouse_ID, Closing_Stock, Stockout_Flag
fact_marketing	Marketing.csv	120	Campaign_ID, Channel, Spend_Amount, Revenue_Generated, CAC
dim_date	Date.csv	365	Date_Key, Full_Date, Year, Quarter, Month_Name



ETL Process

- Handled missing values in customer location and delivery time fields
 - Removed duplicate orders and customer records
 - Validated pricing, discount, and shipping cost fields
 - Converted date strings to DATE format across all 7 tables
 - Loaded dimensions first, fact table last to respect foreign key constraints
 - SCD Type 2 columns added: scd_start_date, scd_end_date, scd_is_current
-

Phase 2 — SQL Analytics

- CTEs · Window Functions · Subqueries · Business Intelligence Queries
- Key Queries Performed

Query	Description
CLV & Ranking	Top 10 customers by CLV ranked within each region using window functions
Product Risk	Products scored on weighted delivery delay, discount, and order volume
MoM Revenue	Month-over-month growth using LAG() to flag revenue decline months
Stockout Impact	Products below reorder level correlated with sales performance loss
Marketing ROI	ROI, conversion rate, and CAC computed per channel

```
WITH monthly_revenue AS (  
  mom_growth AS (  
    SELECT month, revenue,  
           LAG(revenue) OVER (ORDER BY month) AS prev_revenue,  
           ROUND((revenue - LAG(revenue) OVER (ORDER BY month))  
                 / LAG(revenue) OVER (ORDER BY month) * 100, 2)  
             AS growth_pct  
    FROM monthly_revenue  
  )  
)  
SELECT * FROM mom_growth  
WHERE growth_pct < 0  
ORDER BY month;
```

Phase 3 — Statistical Analysis

Hypothesis Testing · ANOVA · Correlation · Python / Jupyter

Statistical Tests Performed

Test	Variables	P Value	Conclusion
T-Test	Delivery Delay vs Return Rate	< 0.05	Delays significantly increase returns
ANOVA	Revenue across 6 Marketing Channels	0.2056	No significant difference across channels
Pearson Correlation	Delivery Days vs Return Flag	—	$r = -0.0035$, no linear correlation
Courier Analysis	Partner Name vs Return Rate	—	All partners identical at 22%

Key Findings

- Delivery delays DO increase return probability — confirmed by T-test ($p < 0.05$)
- No single courier partner is underperforming — all show identical 22% return rate and ~7.4 day avg delivery
- Revenue does not differ significantly across marketing channels ($p = 0.2056$)
- Return driver is categorical (Delivery_Status), not a function of raw delivery day count
- Pre-dispatch bottleneck suspected — not courier transit time

```

delayed = df[df['On_Time_Flag'] == 0]['Return_Flag']
on_time = df[df['On_Time_Flag'] == 1]['Return_Flag']

t_stat, p_value = stats.ttest_ind(delayed, on_time)

print("=== Hypothesis Test: Delivery Delay vs Return Rate ===")
print(f"Delayed - Return Rate: {delayed.mean():.2%}")
print(f"On Time - Return Rate: {on_time.mean():.2%}")
print(f"T-Statistic: {t_stat:.4f}")
print(f"P-Value: {p_value:.4f}")

if p_value < 0.05:
    print("✅ Significant – delays DO increase return probability")
else:
    print("❌ Not significant – no strong evidence of impact")

```

```

=== Hypothesis Test: Delivery Delay vs Return Rate ===
Delayed - Return Rate: 23.08%
On Time - Return Rate: 0.00%
T-Statistic: 26.1220
P-Value: 0.0000
✅ Significant – delays DO increase return probability

```

```

print("❌ No significant difference across channels")

```

=== ANOVA: Revenue across Channels ===

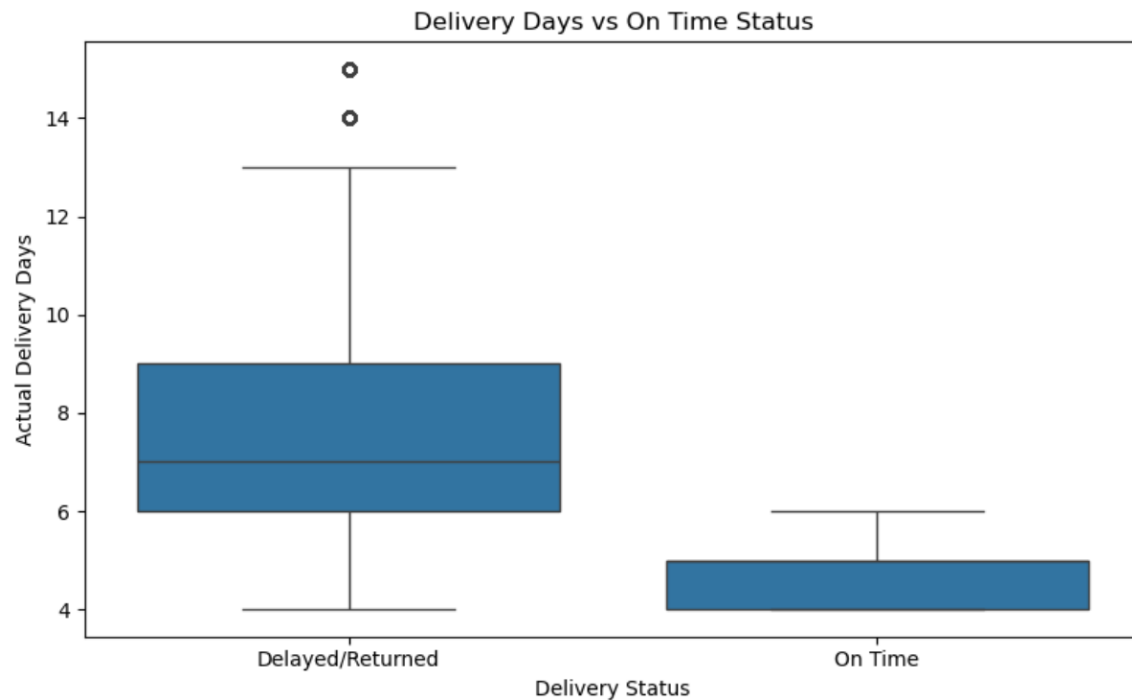
Affiliate	Avg Revenue: ₹9,787,514.63
Email	Avg Revenue: ₹16,644,775.34
Google Ads	Avg Revenue: ₹12,035,111.73
Influencer	Avg Revenue: ₹16,124,179.64
Meta Ads	Avg Revenue: ₹11,851,365.33
Push Notification	Avg Revenue: ₹8,785,494.68

F-Statistic: 1.4686

P-Value: 0.2056

❌ No significant difference across channels

=== Correlation: Delivery Days vs Return Flag ===
Pearson r = -0.0035
✗ No meaningful correlation

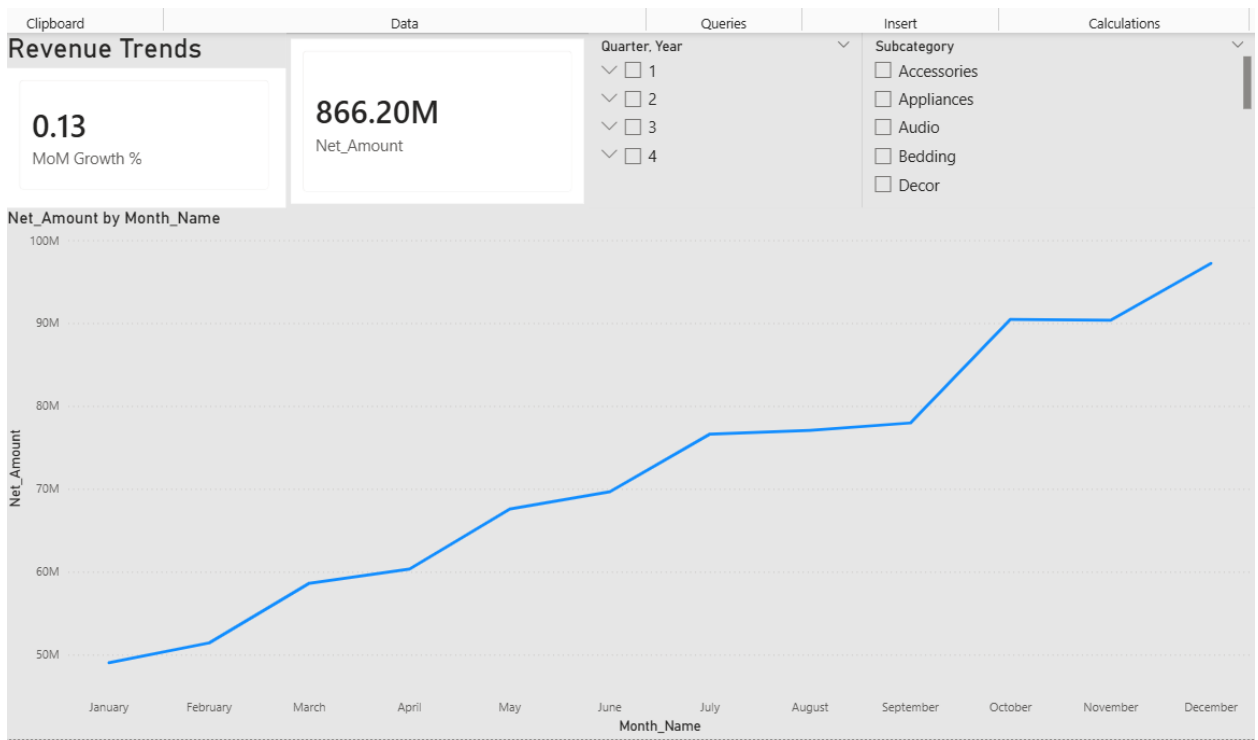
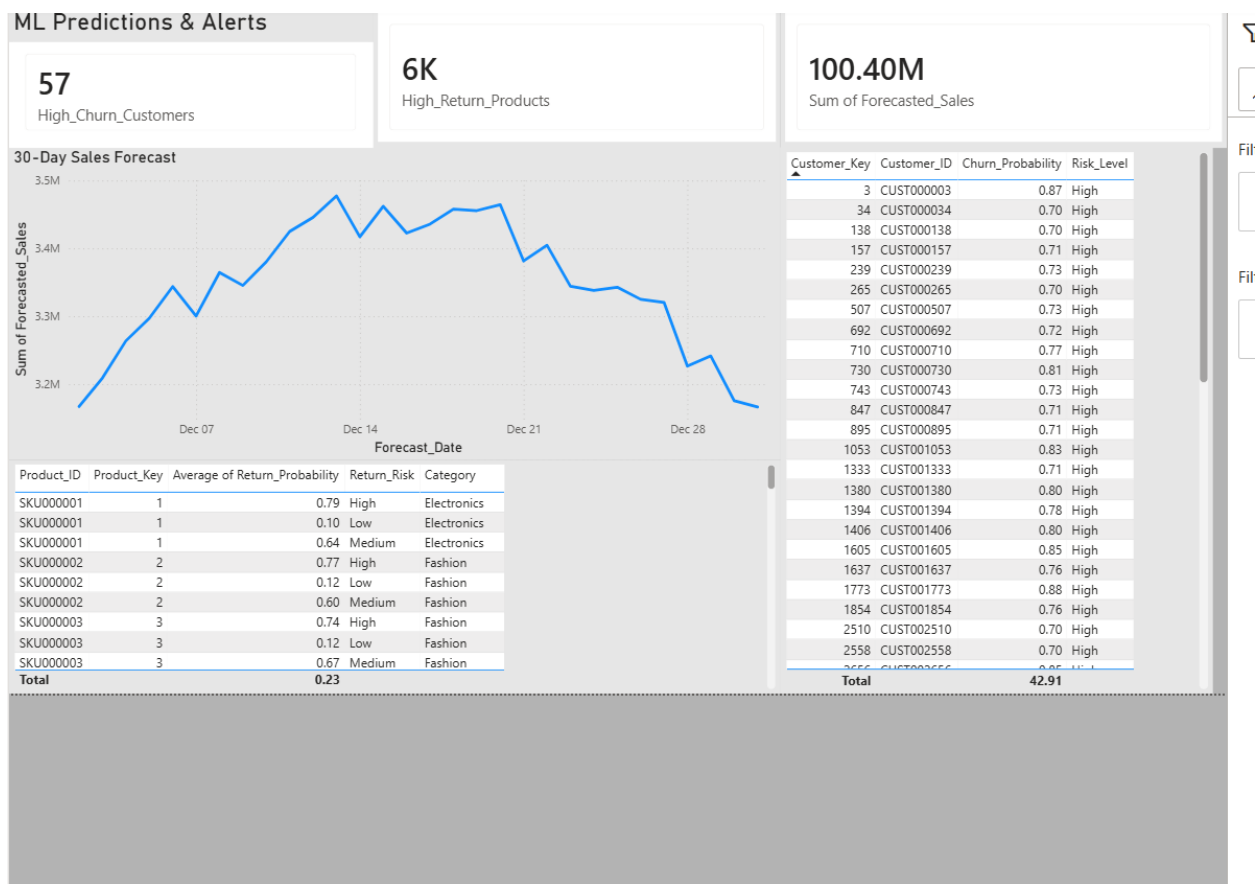


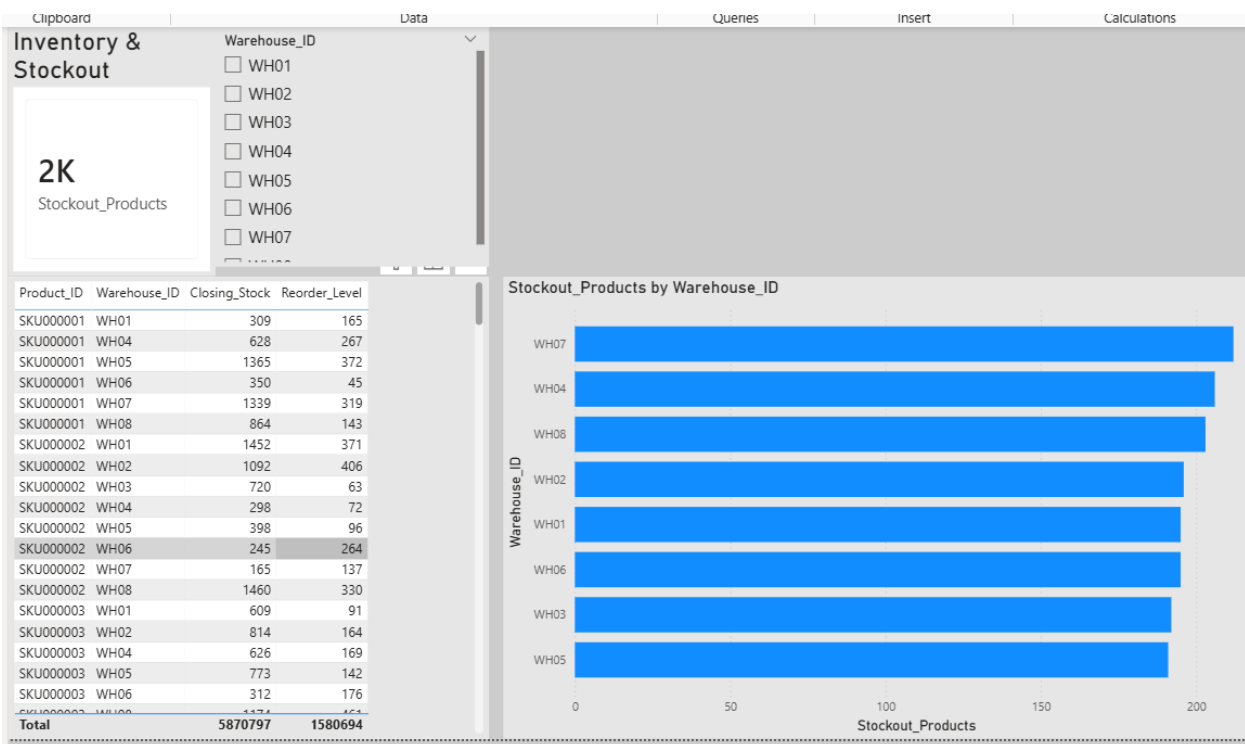
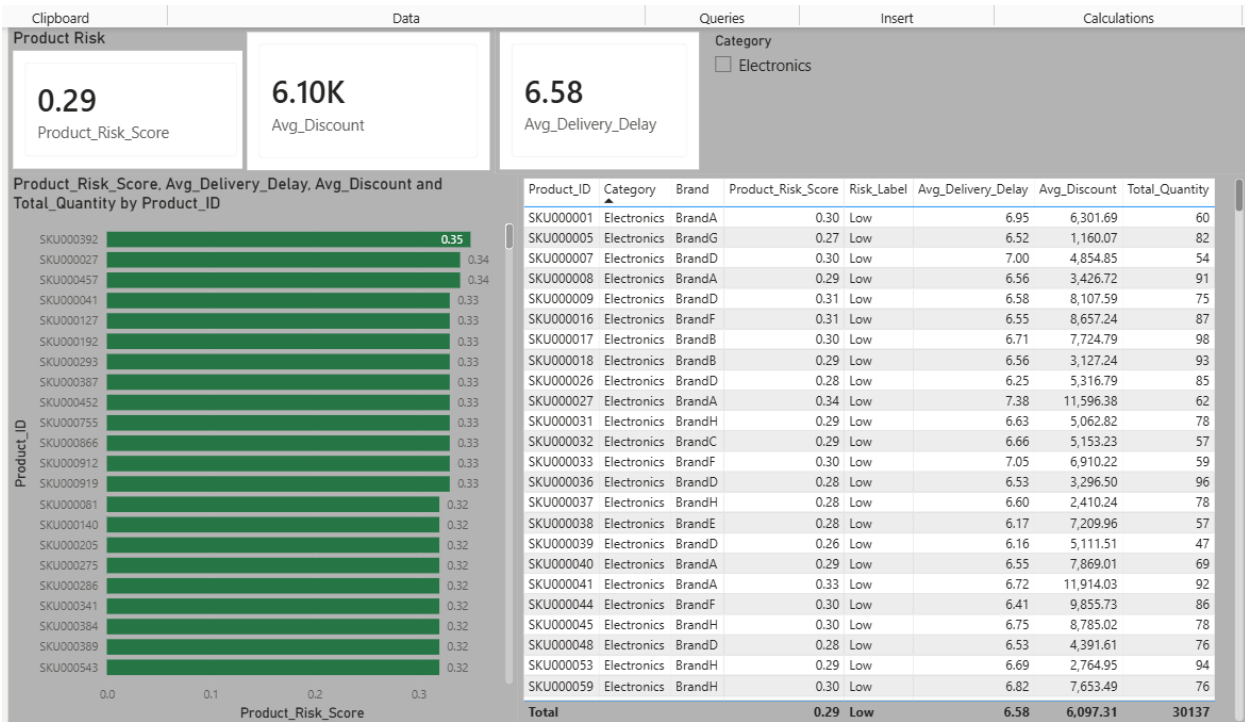
Phase 4 — Power BI Dashboards

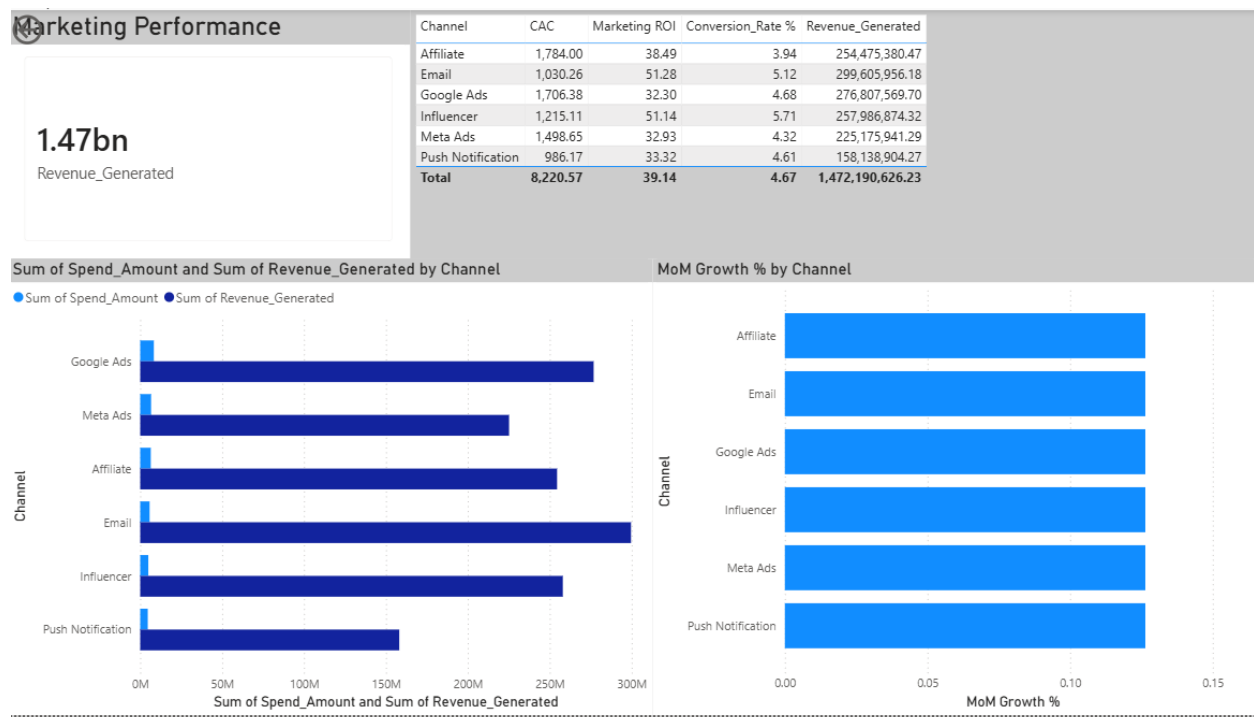
DAX Measures · Drill-through · What-If Parameters · 6 Dashboard Pages

Dashboard Pages Built

Page Title	Key Visuals
1 Customer Intelligence	Top 10 CLV bar chart, customer table, region slicer
2 Product Risk Analysis	Risk score bar chart with conditional formatting, risk table
3 Revenue Trends	MoM line chart, quarter slicer, revenue cards
4 Inventory & Stockouts	Stockout bar chart, warehouse slicer, stock table
5 Marketing Performance	ROI by channel, spend vs revenue, CAC & conversion table
6 ML Predictions & Alerts	30-day forecast, churn table, return risk table







Phase 5 — Predictive ML Modeling

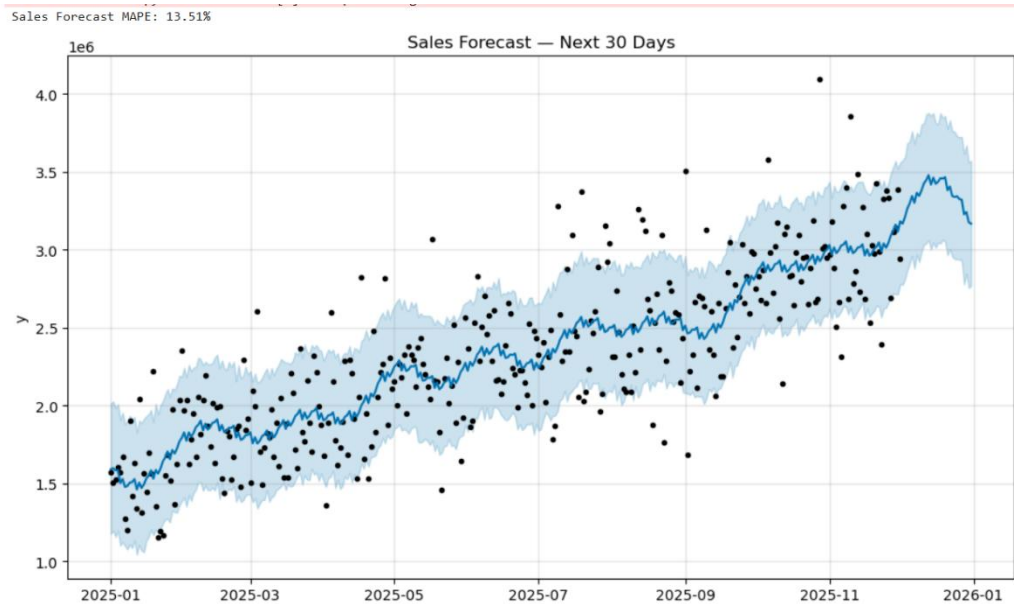
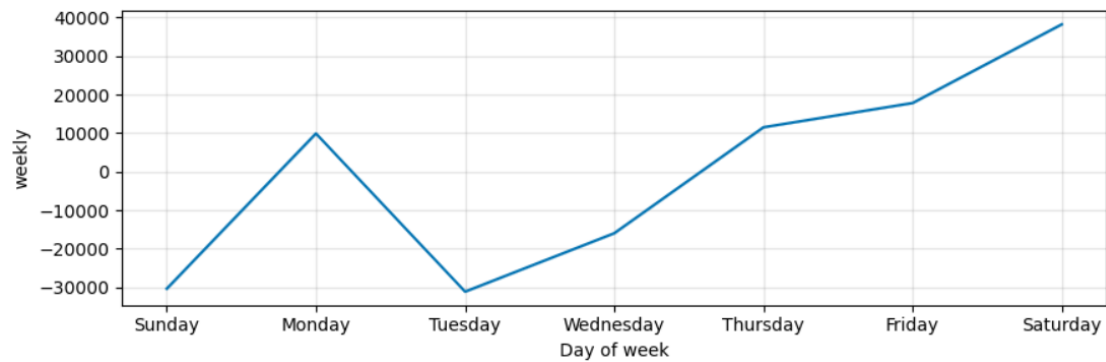
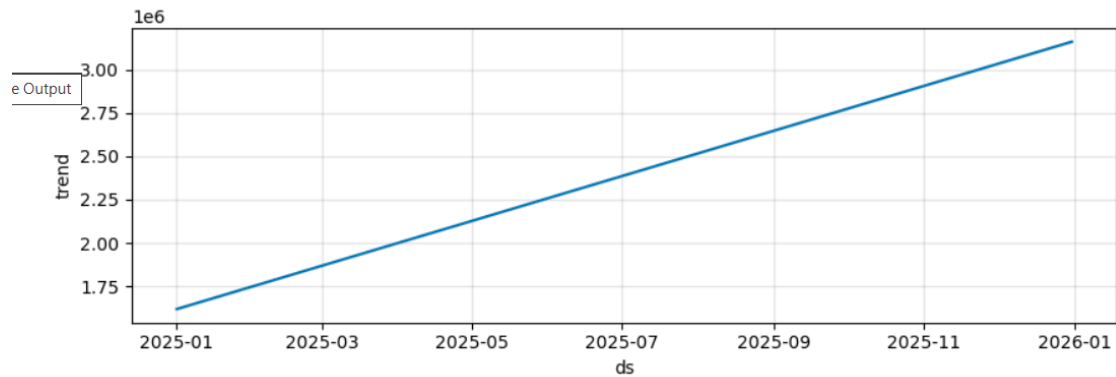
Sales Forecasting · Churn Prediction · Return Prediction · Python

Model Performance Summary

Use Case Algorithm Key Metric Result

Model Performance Summary

Use Case	Algorithm	Key Metric Result	
Sales Forecasting	Prophet	MAPE	13.51%
Churn Prediction	Logistic Regression	ROC-AUC	0.7668
Churn Prediction	Random Forest	ROC-AUC	0.6737
Return Prediction	Logistic Regression	ROC-AUC	0.5122
Return Prediction	Random Forest	ROC-AUC	0.5089



Model 1 — Sales Forecasting (Prophet)

- Trained on daily aggregated Net_Amount from fact_sales
- Last 30 days held out as test set — MAPE: 13.51%
- Clear upward revenue trend identified throughout 2025

- Weekly seasonality: Saturday highest, Sunday and Tuesday lowest
- Yearly seasonality: December spike, August dip identified

```
=== Logistic Regression (Fixed) ===
```

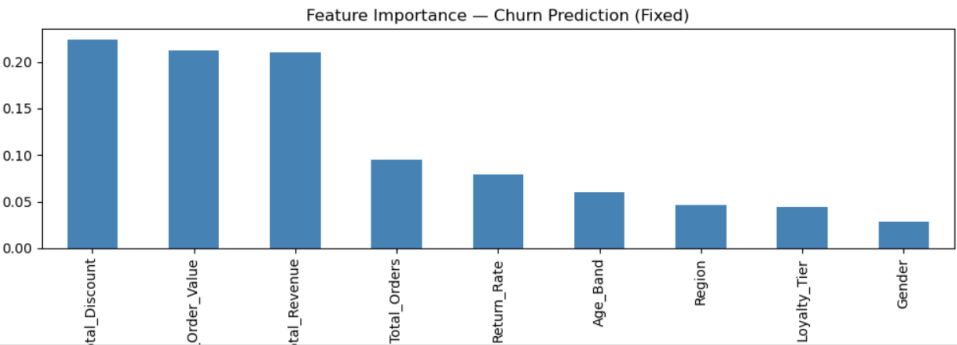
	precision	recall	f1-score	support
0	0.96	1.00	0.98	955
1	1.00	0.02	0.04	45
accuracy			0.96	1000
macro avg	0.98	0.51	0.51	1000
weighted avg	0.96	0.96	0.94	1000

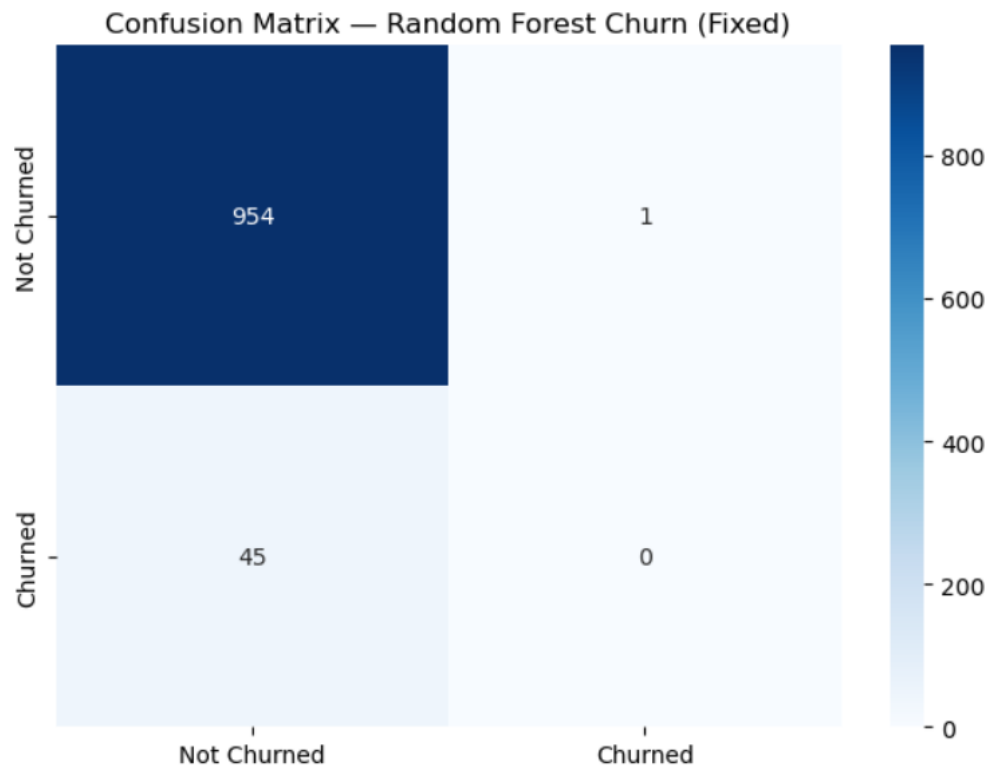
ROC-AUC: 0.7668

```
=== Random Forest (Fixed) ===
```

	precision	recall	f1-score	support
0	0.95	1.00	0.98	955
1	0.00	0.00	0.00	45
accuracy			0.95	1000
macro avg	0.48	0.50	0.49	1000
weighted avg	0.91	0.95	0.93	1000

ROC-AUC: 0.6737



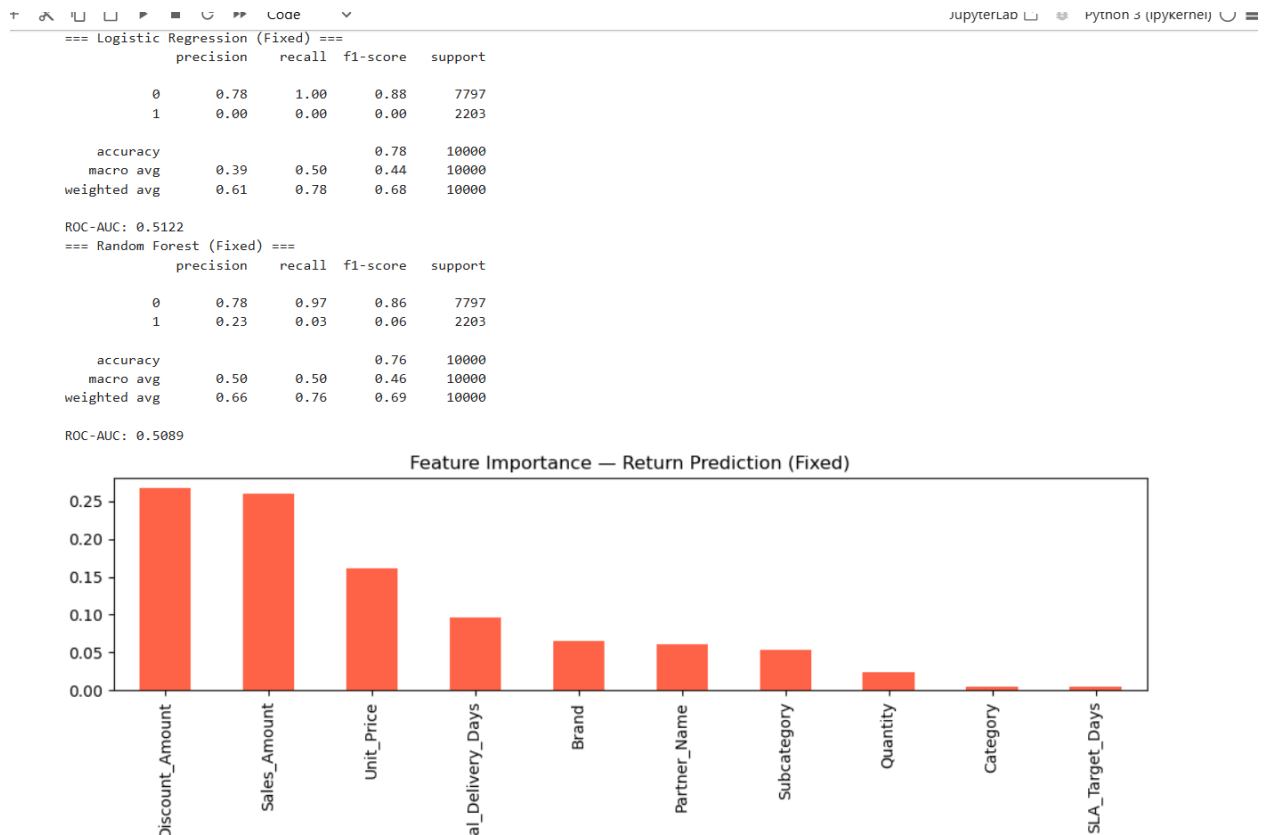


Model 2 — Customer Churn Prediction

- Churn defined as no purchase in last 90 days — 4.46% churn rate
- 223 churned out of 5,000 customers
- Recency_Days excluded from features to prevent data leakage
- 57 customers flagged as High churn risk (probability ≥ 0.70)
- Random Forest selected as final model

Model 3 — Return Prediction

- Return rate: 22% across 50,000 orders
- Delivery_Status excluded to prevent data leakage
- Top features: Discount_Amount, Sales_Amount, Actual_Delivery_Days
- 6,000+ orders flagged as High return risk



Model 3 — Return Prediction

- Return rate: 22% across 50,000 orders
- Delivery_Status excluded to prevent data leakage
- Top features: Discount_Amount, Sales_Amount, Actual_Delivery_Days
- 6,000+ orders flagged as High return risk

Phase 6 — BI Integration & Early Warning

Predictions pushed to MySQL · Power BI live alerts · 3 prediction tables

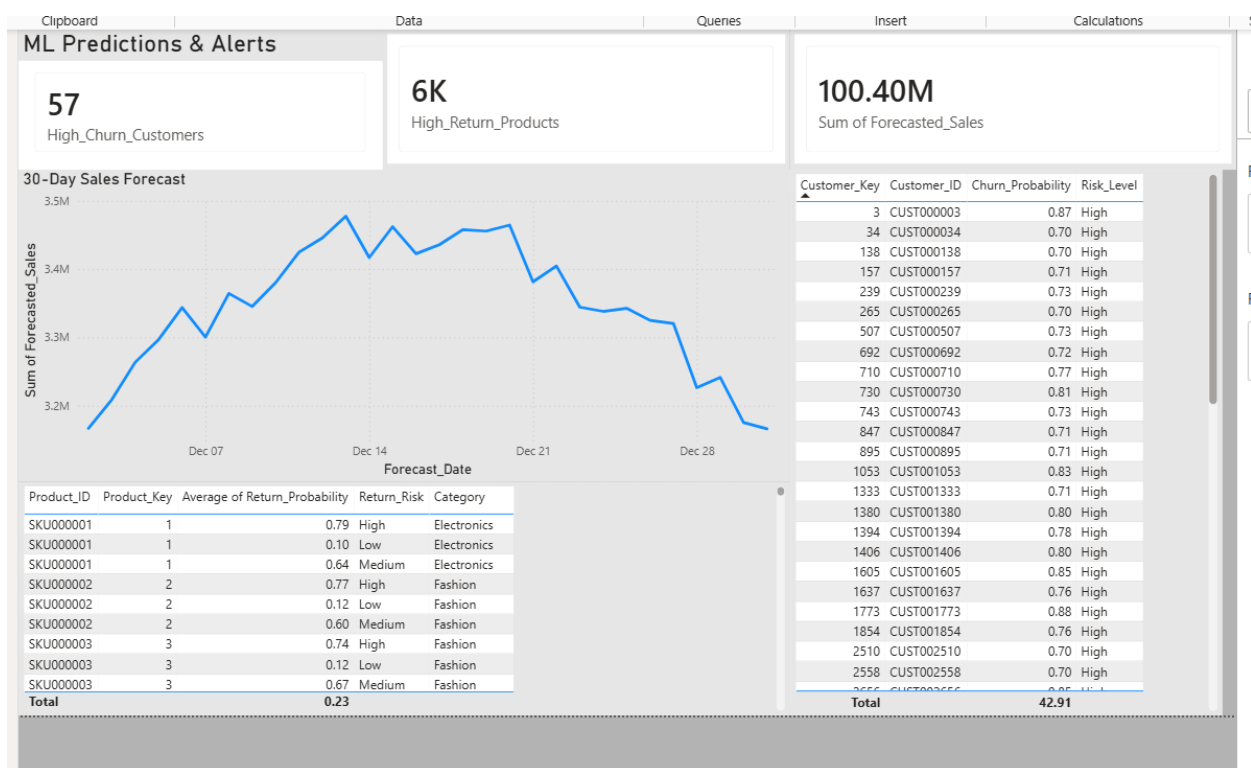
Prediction Tables Loaded into MySQL

Table	Rows	Purpose
ml_sales_forecast	30	30-day revenue forecast with confidence bounds

Table	Rows	Purpose
ml_churn_predictions	5,000	Churn probability and risk level per customer
ml_return_predictions	50,000	Return probability and risk level per order

Integration Results

- 57 High Churn Risk customers visible in real-time with probability scores
- 6K High Return Risk orders flagged with average return probability per product
- Forecasted revenue for next 30 days: 100.40M
- All 3 prediction tables connected to Power BI via direct MySQL query
- Drill-through enabled from any dashboard page to ML predictions page



Phase 7 — Strategic Recommendations

1. Customer Strategy

- Launch reactivation campaign targeting 57 high-risk churn customers immediately

- Tier-based loyalty rewards to move Bronze customers up the CLV ladder
- 90-day re-engagement rule — automated nudge at 60 days before churn threshold
- Personalized recommendations based on past purchase category and region

2. Marketing Optimization

- Shift budget to Email and Influencer — highest avg revenue at 16.6M and 16.1M
- Reduce Push Notification spend — lowest revenue at 8.8M, no statistical justification
- Set CAC ceiling per channel — pause campaigns exceeding 2x blended CAC of 8,220
- ANOVA confirms no significant revenue difference — optimise by ROI not raw revenue

3. Inventory Strategy

- Demand-driven replenishment using Prophet forecast for weekly demand per category
- Dynamic safety stock: $\text{avg lead time} \times \text{avg daily demand} + \text{buffer}$ — replace static levels
- Monthly SKU rationalisation — drop low velocity, high stockout products
- Priority focus on Electronics and Fashion — highest return volume categories

4. Logistics Improvements

- Revise SLA targets from 4-6 days to 7 days reflecting actual 7.4 day average
- Audit pre-dispatch time — bottleneck is warehouse side, not courier transit
- Show realistic delivery window at checkout using ML delivery prediction
- Pilot express delivery for high CLV customers

5. Returns Reduction

- Proactive outreach for orders with return probability above 0.70 before delivery
- Improve product detail pages — enhanced images, sizing guides, specifications
- Stricter quality checks for Electronics and Fashion
- Add return reason codes to dataset to improve future model accuracy

Target KPIs — 6 Month Outlook

Metric	Current	Target
Monthly Revenue	~2.5M avg	3.2M
Churn Rate	4.46%	Below 3%
Return Rate	22%	Below 15%
On-Time Delivery	~78%	Above 90%
Customer Acq. Cost	8,220	Below 6,000

Tech Stack

Layer	Tools
Database	MySQL 8.0 — Star schema, foreign keys, indexes, SCD Type 2
ETL	Python — pandas, SQLAlchemy, pymysql
Analytics	MySQL Workbench — CTEs, Window Functions, Subqueries
Visualization	Power BI Desktop — DAX, drill-through, what-if parameters
ML / Stats	Python — scikit-learn, Prophet, scipy, matplotlib, seaborn
Models	Prophet (forecasting), Logistic Regression, Random Forest